

M.Tech (AIML/DSE) — Machine Learning Assignment 2

Bank Marketing Classification

Problem Statement

Build classification models to predict whether a bank customer will subscribe to a term deposit, evaluate the models using the required metrics, and provide an interactive Streamlit application.

Dataset Description

Bank Marketing — bank-additional-full.csv. The dataset contains 41,188 records, 20 input features and the binary target y (yes/no).

Models and Evaluation

Model	Accuracy	AUC	Precision	Recall	F1	MCC
Logistic Regression	0.9166	0.9424	0.7118	0.4364	0.5411	0.5162
Decision Tree	0.9165	0.9315	0.6613	0.5302	0.5885	0.5468
kNN	0.9082	0.8993	0.6419	0.4192	0.5072	0.4717
Naive Bayes	0.8203	0.8393	0.3495	0.6907	0.4642	0.4009
Random Forest	0.8926	0.9509	0.5135	0.8793	0.6484	0.6199

Overall Winner: Random Forest, based on the highest F1 score (0.6484) in this experiment. It also achieved the highest AUC (0.9509) and MCC (0.6199) among the implemented models.

Observations

Logistic Regression: Strong baseline with good AUC and a reasonable precision/recall balance.

Decision Tree: Captures nonlinear patterns and improves recall compared with Logistic Regression.

kNN: Competitive baseline but affected by the high-dimensional one-hot encoded feature space.

Naive Bayes: High recall but lower precision, reflecting its simplifying independence assumption.

Random Forest: Best overall F1, AUC and MCC in this experiment, giving the strongest balanced performance.

GitHub Repository Link

Replace with your actual GitHub repository URL after upload.

Live Streamlit App Link

Replace with your actual Streamlit Community Cloud URL after deployment.

BITS Virtual Lab Screenshot

Insert the required screenshot showing assignment execution on BITS Virtual Lab before final submission.